Waraitip Laosangprateep

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BKK– Chonburi–Rayong-Samut prakarn · Available from August 2026

PROFILE

Engineering graduate with dual specialisation in Chemical and Medical Engineering, combining expertise in analytical instrumentation, spectroscopic characterisation, and data-driven process analysis. Experienced in designing DOE-style experiments, operating precision optical and electronic instruments, and using Python for data analysis and machine learning. Skilled in root cause analysis and statistical process evaluation. Seeking a technical or application engineering role within a high-technology or precision industrial company.

EDUCATION

Master of Medical Engineering · **University of Auckland** · New Zealand — *Mar 2025 – Jun 2026*

- Thesis: Mechanical characterisation of growth plate structures using microindentation techniques
- Coursework: Advanced Functional Materials · Medical Device & Technology Development · Advanced Drug Disposition & Kinetics · Advanced Imaging Technologies · Engineering Project Management

Bachelor of Engineering – Chemical Engineering · **Thammasat University (SIIT)** · Thailand — *Aug 2020 – May 2024*

- GPA: 3.22 / 4.00 · International Programme, full English instruction
- Senior Project: Theoretical study on butylone inclusion complexes with β-cyclodextrin — presented at PACCON 2024
- Core coursework: Chemical Reaction Engineering · Process Design · Thermodynamics · Heat & Mass Transfer · Fluid Mechanics

RESEARCH & INDUSTRY EXPERIENCE

Master's Thesis Researcher · **University of Auckland** · New Zealand — *Mar 2025 – Present*

- Performed nanoindentation and microindentation mechanical testing on biological tissue samples across growth plate sub-structures
- Applied critical analysis to reconcile conflicting experimental data and form evidence-based conclusions
- Developed expertise in precision instrumentation, sample preparation, and quantitative data interpretation

Research Intern – SWCNTs Defect Characterisation · **Academia Sinica (IAMS-IIP)** · Taiwan — *Jun – Jul 2023*

- Conducted independent experiments on Single-Walled Carbon Nanotubes using absorbance and fluorescence spectroscopy for cancer diagnostic applications
- Designed and executed DOE-style experiments; analysed spectral data and contributed to peer scientific reports
- Gained proficiency in nanomaterial characterisation, precision instrumentation, lab safety protocols, and scientific communication

TECHNICAL SKILLS

Analytical Instrumentation: Absorbance & fluorescence spectroscopy; nanoindentation; optical & electronic instrumentation; scientific instrument operation & calibration

Data & Analytics: Python (data analysis, visualisation, supervised ML classification); statistical analysis; Design of Experiments (DOE); Statistical Process Control (SPC)

Problem-Solving Methods: FMEA (Failure Mode & Effects Analysis); FTA (Fault Tree Analysis); root cause analysis (5-Why · Fishbone); 8D problem solving fundamentals; Lean/Kaizen awareness; data-driven process improvement

Process Simulation: AspenPlus; Gaussian / GaussView; AutoDock; Discovery Studio Visualizer; Origin Lab; WebLabViewer

Engineering Core: Chemical & process engineering; thermodynamics; heat & mass transfer; reaction engineering; fluid mechanics; process scale-up concepts

Quality & Compliance: GMP manufacturing exposure; ISO 13485 / ISO 14971 / IEC 60601 / IEC 62366 (usability engineering) coursework exposure; PHAC certifications (10+); technical report writing

Software & Tools: MS Office Suite (Word, Excel, PowerPoint); CAD (beginner); MS Project; data visualisation tools

ADDITIONAL EXPERIENCE

Customer Service · Auckland, NZ **Lily's Collection · Blue Elephant · Mi&Chi · Khao San Eatery · The Coffee Club** — *Aug 2024 – Jan 2026*

- Developed stakeholder communication, cross-cultural teamwork, and operational reliability skills transferable to customer-site engineering contexts

VOLUNTEER & LEADERSHIP

Narrator & Lab Demonstrator · **Thammasat University – SIIT Open Days** · Thailand — *Jul 2022 – May 2024*

- Facilitated chemical process demonstrations for prospective students; contributed to increased enrolment through clear technical communication

CERTIFICATIONS & ACHIEVEMENTS

- Poster Presentation Certificate – PACCON 2024 (Pure and Applied Chemistry International Conference)
- Internship Certificate – Institute of Atomic and Molecular Sciences, Academia Sinica, Taiwan

LANGUAGES

Thai (Native) · **English** (IELTS 6.5 · TOEIC 830 — Professional) · **Japanese** (Elementary)